

FagusED · A-Level 9709
1.1 Quadratics

What is the central idea in Quadratics?

9709-01-C1

QUESTION

FagusED · A-Level 9709
1.1 Quadratics

Which mechanism, condition or distinction matters in Quadratics?

9709-01-C2

QUESTION

FagusED · A-Level 9709
1.1 Quadratics

Solve $x^2 - 5x + 6 < 0$.

9709-01-P

QUESTION

FagusED · A-Level 9709
1.2 Functions

What is the central idea in Functions?

9709-02-C1

QUESTION

FagusED · A-Level 9709
1.2 Functions

Which mechanism, condition or distinction matters in Functions?

9709-02-C2

QUESTION

FagusED · A-Level 9709
1.2 Functions

For $f(x) = 3x - 2$, find $f^{-1}(x)$ and $f^{-1}(10)$.

9709-02-P

QUESTION

FagusED · A-Level 9709
1.3 Coordinate geometry

What is the central idea in Coordinate geometry?

9709-03-C1

QUESTION

FagusED · A-Level 9709
1.3 Coordinate geometry

Which mechanism, condition or distinction matters in Coordinate geometry?

9709-03-C2

QUESTION

FagusED · A-Level 9709
1.3 Coordinate geometry

Find equation of tangent to circle $x^2 + y^2 = 25$ at (3,4).

9709-03-P

QUESTION

FagusED · A-Level 9709
1.4 Circular measure

What is the central idea in Circular measure?

9709-04-C1

QUESTION

FagusED · A-Level 9709
1.4 Circular measure

Which mechanism, condition or distinction matters in Circular measure?

9709-04-C2

QUESTION

FagusED · A-Level 9709
1.4 Circular measure

A sector has radius 5 cm and angle 1.2 rad. Find arc length and area.

9709-04-P

QUESTION

FagusED · A-Level 9709
1.5 Trigonometry

What is the central idea in Trigonometry?

9709-05-C1

QUESTION

FagusED · A-Level 9709
1.5 Trigonometry

Which mechanism, condition or distinction matters in Trigonometry?

9709-05-C2

QUESTION

FagusED · A-Level 9709
1.5 Trigonometry

Solve $\cos x = -1/2$ for $0 \leq x < 2\pi$.

9709-05-P

QUESTION

FagusED · A-Level 9709
1.6 Series

What is the central idea in Series?

9709-06-C1

QUESTION

FagusED · A-Level 9709
1.6 Series

Which mechanism, condition or distinction matters in Series?

9709-06-C2

QUESTION

FagusED · A-Level 9709
1.6 Series

Find first six terms' sum for geometric series 2, 6, 18, ...

9709-06-P

QUESTION

FagusED · A-Level 9709
1.1 Quadratics

Factor $(x-2)(x-3)<0$; upward parabola negative between roots; $2<x<3$.

9709-01-P

ANSWER

FagusED · A-Level 9709
1.1 Quadratics

A quadratic inequality changes sign at roots when they exist; use a sketch or interval sign test, including endpoints as the question requires.

9709-01-C2

ANSWER

FagusED · A-Level 9709
1.1 Quadratics

Completing the square reveals turning point and range; discriminant b^2-4ac classifies real roots.

9709-01-C1

ANSWER

FagusED · A-Level 9709
1.2 Functions

$y=3x-2$ gives $x=(y+2)/3$; $f^{-1}(10)=4$.

9709-02-P

ANSWER

FagusED · A-Level 9709
1.2 Functions

A composite fg means $f(g(x))$ under common notation, with domain restricted so $g(x)$ lies in f 's domain.

9709-02-C2

ANSWER

FagusED · A-Level 9709
1.2 Functions

A function assigns each allowed input one output. State domain before finding range or inverse; inverse exists on a one-to-one restricted domain.

9709-02-C1

ANSWER

FagusED · A-Level 9709
1.3 Coordinate geometry

Radius gradient $4/3$, tangent gradient $-3/4$; $y-4=-3(x-3)/4$.

9709-03-P

ANSWER

FagusED · A-Level 9709
1.3 Coordinate geometry

A circle $(x-a)^2+(y-b)^2=r^2$ has centre (a,b) . A tangent is perpendicular to radius at contact.

9709-03-C2

ANSWER

FagusED · A-Level 9709
1.3 Coordinate geometry

For a line gradient $m=(y_2-y_1)/(x_2-x_1)$; parallel lines share gradient and perpendicular non-vertical gradients multiply to -1 .

9709-03-C1

ANSWER

FagusED · A-Level 9709
1.4 Circular measure

$s=r\theta=6$ cm; $A=\frac{1}{2}r^2\theta=15$ cm².

9709-04-P

ANSWER

FagusED · A-Level 9709
1.4 Circular measure

Convert θ° to $\theta\pi/180$; for a segment subtract triangle area $\frac{1}{2}r^2\sin\theta$ from sector area.

9709-04-C2

ANSWER

FagusED · A-Level 9709
1.4 Circular measure

In radians, arc length $s=r\theta$ and sector area $A=\frac{1}{2}r^2\theta$. These formulas require θ in radians, not degrees.

9709-04-C1

ANSWER

FagusED · A-Level 9709
1.5 Trigonometry

Reference angle $\pi/3$; cosine negative in quadrants II and III; $x=2\pi/3, 4\pi/3$.

9709-05-P

ANSWER

FagusED · A-Level 9709
1.5 Trigonometry

For trig equations over an interval, find all solutions using symmetry and periodicity; do not accept inverse-calculator principal value alone.

9709-05-C2

ANSWER

FagusED · A-Level 9709
1.5 Trigonometry

Use exact unit-circle values and signs by quadrant. Identities $\sin^2x+\cos^2x=1$ and $\tan x=\sin x/\cos x$ simplify expressions.

9709-05-C1

ANSWER

FagusED · A-Level 9709
1.6 Series

$a=2, r=3$; $S_6=2(3^6-1)/(3-1)=728$.

9709-06-P

ANSWER

FagusED · A-Level 9709
1.6 Series

A geometric sequence has $u_n=ar^{n-1}$, $S_n=a(1-r^n)/(1-r)$ if $r\neq 1$, and infinite sum $a/(1-r)$ only for $\text{abs}(r)<1$.

9709-06-C2

ANSWER

FagusED · A-Level 9709
1.6 Series

Arithmetic sequences have $u_n=a+(n-1)d$ and $S_n=n[2a+(n-1)d]/2$.

9709-06-C1

ANSWER

FagusED · A-Level 9709
1.7 Differentiation

What is the central idea in Differentiation?

9709-07-C1

QUESTION

FagusED · A-Level 9709
1.7 Differentiation

Which mechanism, condition or distinction matters in Differentiation?

9709-07-C2

QUESTION

FagusED · A-Level 9709
1.7 Differentiation

Differentiate $y=(2x+1)^3$ and find gradient at $x=1$.

9709-07-P

QUESTION

FagusED · A-Level 9709
1.8 Integration

What is the central idea in Integration?

9709-08-C1

QUESTION

FagusED · A-Level 9709
1.8 Integration

Which mechanism, condition or distinction matters in Integration?

9709-08-C2

QUESTION

FagusED · A-Level 9709
1.8 Integration

Evaluate $\int$ from 1 to 3 of $2x \, dx$.

9709-08-P

QUESTION

FagusED · A-Level 9709
2.1 Algebra

What is the central idea in Algebra?

9709-09-C1

QUESTION

FagusED · A-Level 9709
2.1 Algebra

Which mechanism, condition or distinction matters in Algebra?

9709-09-C2

QUESTION

FagusED · A-Level 9709
2.1 Algebra

Expand $(1+x)^{-2}$ to terms in x^2 and state validity.

9709-09-P

QUESTION

FagusED · A-Level 9709
2.2 Logarithmic and exponential functions

What is the central idea in Logarithmic and exponential functions?

9709-10-C1

QUESTION

FagusED · A-Level 9709
2.2 Logarithmic and exponential functions

Which mechanism, condition or distinction matters in Logarithmic and exponential functions?

9709-10-C2

QUESTION

FagusED · A-Level 9709
2.2 Logarithmic and exponential functions

Solve $\ln(x-1)=2$ for real x .

9709-10-P

QUESTION

FagusED · A-Level 9709
2.3 Trigonometry

What is the central idea in Trigonometry?

9709-11-C1

QUESTION

FagusED · A-Level 9709
2.3 Trigonometry

Which mechanism, condition or distinction matters in Trigonometry?

9709-11-C2

QUESTION

FagusED · A-Level 9709
2.3 Trigonometry

Solve $\sin 2x=0$ for $0 \leq x < 2\pi$.

9709-11-P

QUESTION

FagusED · A-Level 9709
1.7 Differentiation

Chain rule $y' = 3(2x+1)^2 \times 2 = 6(2x+1)^2$; at $x=1$ gradient 54.

9709-07-P

ANSWER

FagusED · A-Level 9709
1.7 Differentiation

Stationary points satisfy derivative zero, but classify with second derivative or sign change. Related rates require units and sign.

9709-07-C2

ANSWER

FagusED · A-Level 9709
1.7 Differentiation

Derivative is instantaneous gradient and rate of change. Power rule $d(x^n)/dx = nx^{n-1}$; use product, quotient and chain rules where needed.

9709-07-C1

ANSWER

FagusED · A-Level 9709
1.8 Integration

Antiderivative x^2 ; x^2 evaluated 1 to 3 $= 9 - 1 = 8$.

9709-08-P

ANSWER

FagusED · A-Level 9709
1.8 Integration

A substitution reverses the chain rule and usually changes integration limits in a definite integral.

9709-08-C2

ANSWER

FagusED · A-Level 9709
1.8 Integration

Integration reverses differentiation up to a constant C. For a definite integral use antiderivative limits; signed area below x-axis is negative.

9709-08-C1

ANSWER

FagusED · A-Level 9709
2.1 Algebra

Use binomial coefficients: $1 - 2x + 3x^2$; $\text{abs}(x) < 1$.

9709-09-P

ANSWER

FagusED · A-Level 9709
2.1 Algebra

Partial fractions simplify rational integration; first factor denominator and match numerator degrees. Factor theorem provides roots and polynomial division reduces degree.

9709-09-C2

ANSWER

FagusED · A-Level 9709
2.1 Algebra

Binomial expansion for non-integer powers is infinite and valid only for $\text{abs}(x) < 1$ in $(1+x)^n$.

9709-09-C1

ANSWER

FagusED · A-Level 9709
2.2 Logarithmic and exponential functions

Domain $x > 1$; exponentiate: $x - 1 = e^2$; $x = 1 + e^2$.

9709-10-P

ANSWER

FagusED · A-Level 9709
2.2 Logarithmic and exponential functions

Exponential growth/decay solves $dy/dt = ky$ and requires the initial condition to determine the multiplier.

9709-10-C2

ANSWER

FagusED · A-Level 9709
2.2 Logarithmic and exponential functions

ln and exp are inverses on appropriate domains. Log rules combine products into sums and powers into coefficients; do not split $\ln(a+b)$.

9709-10-C1

ANSWER

FagusED · A-Level 9709
2.3 Trigonometry

$2x = n\pi$; $x = n\pi/2$; $x = 0, \pi/2, \pi, 3\pi/2$.

9709-11-P

ANSWER

FagusED · A-Level 9709
2.3 Trigonometry

State the full interval after algebra and test solutions excluded by any division by $\sin x$ or $\cos x$.

9709-11-C2

ANSWER

FagusED · A-Level 9709
2.3 Trigonometry

At this level, double-angle identities $\sin 2x = 2 \sin x \cos x$ and $\cos 2x = 2 \cos^2 x - 1$ help solve equations and integrate.

9709-11-C1

ANSWER

FagusED · A-Level 9709
2.4 Differentiation

What is the central idea in Differentiation?

9709-12-C1

QUESTION

FagusED · A-Level 9709
2.4 Differentiation

Which mechanism, condition or distinction matters in Differentiation?

9709-12-C2

QUESTION

FagusED · A-Level 9709
2.4 Differentiation

For $x=t^2+1$, $y=t^3$, calculate dy/dx at $t=2$.

9709-12-P

QUESTION

FagusED · A-Level 9709
2.5 Integration

What is the central idea in Integration?

9709-13-C1

QUESTION

FagusED · A-Level 9709
2.5 Integration

Which mechanism, condition or distinction matters in Integration?

9709-13-C2

QUESTION

FagusED · A-Level 9709
2.5 Integration

Evaluate $\int$ from 0 to 1 of $2x(x^2+1)$ dx.

9709-13-P

QUESTION

FagusED · A-Level 9709
2.6 Numerical solution of equations

What is the central idea in Numerical solution of equations?

9709-14-C1

QUESTION

FagusED · A-Level 9709
2.6 Numerical solution of equations

Which mechanism, condition or distinction matters in Numerical solution of equations?

9709-14-C2

QUESTION

FagusED · A-Level 9709
2.6 Numerical solution of equations

Show $f(x)=x^3-2$ has a root between 1 and 2 and perform one Newton step from $x_0=1.5$.

9709-14-P

QUESTION

FagusED · A-Level 9709
3.1 Algebra

What is the central idea in Algebra?

9709-15-C1

QUESTION

FagusED · A-Level 9709
3.1 Algebra

Which mechanism, condition or distinction matters in Algebra?

9709-15-C2

QUESTION

FagusED · A-Level 9709
3.1 Algebra

Decompose $3/[x(x+3)]$ into simple fractions.

9709-15-P

QUESTION

FagusED · A-Level 9709
3.2 Logarithmic and exponential functions

What is the central idea in Logarithmic and exponential functions?

9709-16-C1

QUESTION

FagusED · A-Level 9709
3.2 Logarithmic and exponential functions

Which mechanism, condition or distinction matters in Logarithmic and exponential functions?

9709-16-C2

QUESTION

FagusED · A-Level 9709
3.2 Logarithmic and exponential functions

Solve $e^{(2x)}=7$ and give exact x.

9709-16-P

QUESTION

$dy/dt=3t^2$, $dx/dt=2t$; $dy/dx=3t/2$;
at 2 gradient 3.

9709-12-P

ANSWER

Set $dy/dx=0$ carefully because a zero numerator with zero denominator may need separate analysis.

9709-12-C2

ANSWER

Implicit differentiation differentiates y as $y(x)$, introducing dy/dx . Parametric curves use $dy/dx=(dy/dt)/(dx/dt)$; chain and product rules handle composite products.

9709-12-C1

ANSWER

Substitute $u=x^2+1$, $du=2x dx$;
integrate u from 1 to 2;
 $[u^2/2]_1^2=3/2$.

9709-13-P

ANSWER

Area of a region between graphs is integral of upper minus lower curve; split at intersections. For volumes of revolution about an axis use $\pi y^2 dx$ when permitted by syllabus.

9709-13-C2

ANSWER

Integration by substitution identifies an inner expression whose derivative also occurs.

9709-13-C1

ANSWER

$f(1)=-1$, $f(2)=6$ so a root lies between; $f'=3x^2$; $x_1=1.5-(1.5^3-2)/(3 \times 1.5^2) \approx 1.2963$.

9709-14-P

ANSWER

Iteration $x_{n+1}=g(x_n)$ is useful if it converges; convergence near a fixed point is favoured when $\text{abs}(g'(x)) < 1$. Report a final answer only after checking required accuracy.

9709-14-C2

ANSWER

A sign change $f(a)f(b) < 0$ for continuous f brackets a root, subject to continuity.

9709-14-C1

ANSWER

Set $A/x+B/(x+3)$: $3=A(x+3)+Bx$;
 $x=0$ gives $A=1$, $x=-3$ gives $B=-1$.

9709-15-P

ANSWER

For a rational expression, divide first if numerator degree is not lower than denominator, then decompose using factors and repeated-factor terms. Validate by recombining or testing convenient x .

9709-15-C2

ANSWER

Advanced algebra includes factorisation, partial fractions and binomial series as appropriate.

9709-15-C1

ANSWER

Take logs: $2x=\ln 7$; $x=(\ln 7)/2$.

9709-16-P

ANSWER

Differentiation
 $d(e^{(kx)})/dx = ke^{(kx)}$;
 $d(\ln \text{abs}(x))/dx = 1/x$ for $x \neq 0$. Check original domain after solving logarithmic expressions.

9709-16-C2

ANSWER

Use logarithms to linearise exponential equations and transform data when a model $y=Ae^{(kx)}$ is justified.

9709-16-C1

ANSWER

FagusED · A-Level 9709
3.3 Trigonometry

What is the central idea in Trigonometry?

9709-17-C1

QUESTION

FagusED · A-Level 9709
3.3 Trigonometry

Which mechanism, condition or distinction matters in Trigonometry?

9709-17-C2

QUESTION

FagusED · A-Level 9709
3.3 Trigonometry

Write $5\cos x + 12\sin x$ as $R\cos(x-\alpha)$ with $R>0$ and α in first quadrant.

9709-17-P

QUESTION

FagusED · A-Level 9709
3.4 Differentiation

What is the central idea in Differentiation?

9709-18-C1

QUESTION

FagusED · A-Level 9709
3.4 Differentiation

Which mechanism, condition or distinction matters in Differentiation?

9709-18-C2

QUESTION

FagusED · A-Level 9709
3.4 Differentiation

Differentiate $y = x e^{(2x)}$ and find y' at $x=0$.

9709-18-P

QUESTION

FagusED · A-Level 9709
3.5 Integration

What is the central idea in Integration?

9709-19-C1

QUESTION

FagusED · A-Level 9709
3.5 Integration

Which mechanism, condition or distinction matters in Integration?

9709-19-C2

QUESTION

FagusED · A-Level 9709
3.5 Integration

Evaluate $\int$ from 0 to 1 of $1/(x+1)$ dx.

9709-19-P

QUESTION

FagusED · A-Level 9709
3.6 Numerical solution of equations

What is the central idea in Numerical solution of equations?

9709-20-C1

QUESTION

FagusED · A-Level 9709
3.6 Numerical solution of equations

Which mechanism, condition or distinction matters in Numerical solution of equations?

9709-20-C2

QUESTION

FagusED · A-Level 9709
3.6 Numerical solution of equations

For $f(x)=x^2-3$ and $x_0=2$, calculate one Newton step.

9709-20-P

QUESTION

FagusED · A-Level 9709
3.7 Vectors

What is the central idea in Vectors?

9709-21-C1

QUESTION

FagusED · A-Level 9709
3.7 Vectors

Which mechanism, condition or distinction matters in Vectors?

9709-21-C2

QUESTION

FagusED · A-Level 9709
3.7 Vectors

For $a=(1,2,2)$ and $b=(2,0,0)$, find $\cos\theta$ between them.

9709-21-P

QUESTION

FagusED · A-Level 9709
3.3 Trigonometry

$R = \sqrt{25+144}=13$; $R\cos\alpha=5$,
 $R\sin\alpha=12$; expression
 $13\cos(x-\arctan(12/5))$.

9709-17-P

ANSWER

FagusED · A-Level 9709
3.3 Trigonometry

For equations, list every solution over the specified interval and reject extraneous values from squaring.

9709-17-C2

ANSWER

FagusED · A-Level 9709
3.3 Trigonometry

Use addition formulae, double-angle identities and $R\cos(x-\alpha)$ transformations. A transformed form reveals amplitude and phase; choose α using both sine and cosine coefficients.

9709-17-C1

ANSWER

FagusED · A-Level 9709
3.4 Differentiation

Product and chain: $y' = e^{(2x)} + 2xe^{(2x)} = e^{(2x)}(1+2x)$; at $x=0, 1$.

9709-18-P

ANSWER

FagusED · A-Level 9709
3.4 Differentiation

Implicit/parametric second derivatives require care: $d^2y/dx^2 = (d/dt(dy/dx))/(dx/dt)$. Identify stationary points and inflection using derivatives and sign tests.

9709-18-C2

ANSWER

FagusED · A-Level 9709
3.4 Differentiation

Differentiate composite trig, exponential and logarithmic functions with chain, product and quotient rules.

9709-18-C1

ANSWER

FagusED · A-Level 9709
3.5 Integration

Antiderivative $\ln \text{abs}(x+1)$;
 $\ln 2 - \ln 1 = \ln 2$.

9709-19-P

ANSWER

FagusED · A-Level 9709
3.5 Integration

For integrals with a linear inner function multiply by reciprocal of its derivative. Improper integrals need limiting processes when bounds or integrand are unbounded.

9709-19-C2

ANSWER

FagusED · A-Level 9709
3.5 Integration

Advanced methods include partial fractions, substitution and parts, and trig/exponential integrals.

9709-19-C1

ANSWER

FagusED · A-Level 9709
3.6 Numerical solution of equations

$f(2)=1$, $f'(2)=4$; $x_1=2-1/4=1.75$.

9709-20-P

ANSWER

FagusED · A-Level 9709
3.6 Numerical solution of equations

Interval bisection gives a guaranteed bracket under continuity and sign change. Numerical answers require a reason for chosen rounding and a check at boundaries.

9709-20-C2

ANSWER

FagusED · A-Level 9709
3.6 Numerical solution of equations

Newton-Raphson updates $x_{n+1} = x_n - f(x_n)/f'(x_n)$; if f' is small it may jump far or fail.

9709-20-C1

ANSWER

FagusED · A-Level 9709
3.7 Vectors

$a \cdot b = 2$, $\text{abs}(a)=3$ and $\text{abs}(b)=2$;
 $\cos\theta = 2/6 = 1/3$.

9709-21-P

ANSWER

FagusED · A-Level 9709
3.7 Vectors

To intersect two lines, solve all three component equations for their separate parameters; a pairwise apparent intersection may be skew if the third fails.

9709-21-C2

ANSWER

FagusED · A-Level 9709
3.7 Vectors

A line in 3D has $r = a + \lambda b$. Dot product $a \cdot b = \text{abs}(a)\text{abs}(b)\cos\theta$ gives angle and perpendicularity.

9709-21-C1

ANSWER

FagusED · A-Level 9709
3.8 Differential equations

What is the central idea in Differential equations?

9709-22-C1

QUESTION

FagusED · A-Level 9709
3.8 Differential equations

Which mechanism, condition or distinction matters in Differential equations?

9709-22-C2

QUESTION

FagusED · A-Level 9709
3.8 Differential equations

Solve $dy/dx = -0.5y$ with $y(0)=8$.

9709-22-P

QUESTION

FagusED · A-Level 9709
3.9 Complex numbers

What is the central idea in Complex numbers?

9709-23-C1

QUESTION

FagusED · A-Level 9709
3.9 Complex numbers

Which mechanism, condition or distinction matters in Complex numbers?

9709-23-C2

QUESTION

FagusED · A-Level 9709
3.9 Complex numbers

Express $(3+4i)/(1-i)$ in $a+bi$ form.

9709-23-P

QUESTION

FagusED · A-Level 9709
4.1 Forces and equilibrium

What is the central idea in Forces and equilibrium?

9709-24-C1

QUESTION

FagusED · A-Level 9709
4.1 Forces and equilibrium

Which mechanism, condition or distinction matters in Forces and equilibrium?

9709-24-C2

QUESTION

FagusED · A-Level 9709
4.1 Forces and equilibrium

A block weighs 20 N on horizontal surface and is pulled horizontally by 6 N while at rest. State friction and reaction, assuming no other vertical forces.

9709-24-P

QUESTION

FagusED · A-Level 9709
4.2 Kinematics of motion in a straight line

What is the central idea in Kinematics of motion in a straight line?

9709-25-C1

QUESTION

FagusED · A-Level 9709
4.2 Kinematics of motion in a straight line

Which mechanism, condition or distinction matters in Kinematics of motion in a straight line?

9709-25-C2

QUESTION

FagusED · A-Level 9709
4.2 Kinematics of motion in a straight line

A particle has $v=4t-2$ m s⁻¹ and $s(0)=3$ m. Find $s(t)$ and displacement from $t=0$ to 2.

9709-25-P

QUESTION

FagusED · A-Level 9709
4.3 Momentum

What is the central idea in Momentum?

9709-26-C1

QUESTION

FagusED · A-Level 9709
4.3 Momentum

Which mechanism, condition or distinction matters in Momentum?

9709-26-C2

QUESTION

FagusED · A-Level 9709
4.3 Momentum

Two 0.5 kg carts, one at 6 m s⁻¹ and one stationary, stick. Find speed and lost kinetic energy.

9709-26-P

QUESTION

FagusED · A-Level 9709
3.8 Differential equations

Separate $dy/y = -0.5 dx$;
 $\ln abs(y) = -0.5x + C$;
 $y = 8e^{(-0.5x)}$.

9709-22-P

ANSWER

FagusED · A-Level 9709
3.8 Differential equations

Model assumptions determine meaningful domain; initial conditions select a particular solution. Exponential change arises when rate is proportional to current amount.

9709-22-C2

ANSWER

FagusED · A-Level 9709
3.8 Differential equations

Separable differential equations rearrange to integrate functions of y and x on opposite sides.

9709-22-C1

ANSWER

FagusED · A-Level 9709
3.9 Complex numbers

Multiply top and bottom by $(1+i)$;
numerator $(3+4i)(1+i) = -1+7i$;
divide 2: $-1/2 + (7/2)i$.

9709-23-P

ANSWER

FagusED · A-Level 9709
3.9 Complex numbers

Real-coefficient polynomial equations have complex roots in conjugate pairs, a fact useful when solving quadratics and higher-degree equations.

9709-23-C2

ANSWER

FagusED · A-Level 9709
3.9 Complex numbers

Write $z = x + iy$; modulus and argument give polar form. Multiplication adds arguments and multiplies moduli; conjugate multiplication rationalises division.

9709-23-C1

ANSWER

FagusED · A-Level 9709
4.1 Forces and equilibrium

Friction = 6 N opposite pull;
normal reaction = 20 N.

9709-24-P

ANSWER

FagusED · A-Level 9709
4.1 Forces and equilibrium

Friction is bounded by μR for limiting friction; otherwise its actual magnitude is determined by equilibrium/dynamics, not automatically μR .

9709-24-C2

ANSWER

FagusED · A-Level 9709
4.1 Forces and equilibrium

Draw a free-body diagram and resolve each force into horizontal and vertical components. Equilibrium requires zero resultant in both directions.

9709-24-C1

ANSWER

FagusED · A-Level 9709
4.2 Kinematics of motion in a straight line

Integrate: $s = 2t^2 - 2t + C$, $C = 3$;
 $s(2) - s(0) = 4$ m.

9709-25-P

ANSWER

FagusED · A-Level 9709
4.2 Kinematics of motion in a straight line

Constant-acceleration equations require truly constant a ; sign conventions must be maintained in multi-stage travel.

9709-25-C2

ANSWER

FagusED · A-Level 9709
4.2 Kinematics of motion in a straight line

Velocity is derivative of displacement; acceleration is derivative of velocity. Integrate acceleration to velocity and velocity to displacement with constants fixed by conditions.

9709-25-C1

ANSWER

FagusED · A-Level 9709
4.3 Momentum

Momentum gives $v = (0.5 \times 6)/1 = 3$ m s^{-1} ; KE initial 9 J, final 4.5 J, loss 4.5 J.

9709-26-P

ANSWER

FagusED · A-Level 9709
4.3 Momentum

Kinetic energy may decrease even when momentum is conserved, especially when bodies stick.

9709-26-C2

ANSWER

FagusED · A-Level 9709
4.3 Momentum

Linear momentum $p = mv$ has direction; impulse $F\Delta t$ changes momentum. For collisions, total momentum is conserved if external impulse is negligible.

9709-26-C1

ANSWER

FagusED · A-Level 9709
4.4 Newton's laws of motion

What is the central idea in Newton's laws of motion?

9709-27-C1

QUESTION

FagusED · A-Level 9709
4.4 Newton's laws of motion

Which mechanism, condition or distinction matters in Newton's laws of motion?

9709-27-C2

QUESTION

FagusED · A-Level 9709
4.4 Newton's laws of motion

A 4 kg mass hangs against a 1 kg mass over a smooth pulley, $g=10$. Find acceleration.

9709-27-P

QUESTION

FagusED · A-Level 9709
4.5 Energy, work and power

What is the central idea in Energy, work and power?

9709-28-C1

QUESTION

FagusED · A-Level 9709
4.5 Energy, work and power

Which mechanism, condition or distinction matters in Energy, work and power?

9709-28-C2

QUESTION

FagusED · A-Level 9709
4.5 Energy, work and power

A 2 kg object falls 5 m from rest without resistance, $g=10$. Determine speed using energy conservation.

9709-28-P

QUESTION

FagusED · A-Level 9709
5.1 Representation of data

What is the central idea in Representation of data?

9709-29-C1

QUESTION

FagusED · A-Level 9709
5.1 Representation of data

Which mechanism, condition or distinction matters in Representation of data?

9709-29-C2

QUESTION

FagusED · A-Level 9709
5.1 Representation of data

Two histogram classes have frequencies 12 and 18 with widths 3 and 6. Find both frequency densities.

9709-29-P

QUESTION

FagusED · A-Level 9709
5.2 Permutations and combinations

What is the central idea in Permutations and combinations?

9709-30-C1

QUESTION

FagusED · A-Level 9709
5.2 Permutations and combinations

Which mechanism, condition or distinction matters in Permutations and combinations?

9709-30-C2

QUESTION

FagusED · A-Level 9709
5.2 Permutations and combinations

How many different arrangements of letters LEVEL are there?

9709-30-P

QUESTION

FagusED · A-Level 9709
5.3 Probability

What is the central idea in Probability?

9709-31-C1

QUESTION

FagusED · A-Level 9709
5.3 Probability

Which mechanism, condition or distinction matters in Probability?

9709-31-C2

QUESTION

FagusED · A-Level 9709
5.3 Probability

A bag has three blue and two green counters; draw two without replacement. Find probability of one of each.

9709-31-P

QUESTION

FagusED · A-Level 9709
4.4 Newton's laws of motion

Driving force difference
 $(4-1)g=30\text{ N}$; total mass 5 kg ;
 $a=6\text{ m s}^{-2}$ toward 4 kg side.

9709-27-P

ANSWER

FagusED · A-Level 9709
4.4 Newton's laws of motion

In connected-particle models,
write an equation for each body
and use a common acceleration;
tension is an internal interaction
and cancels when treating both as
one system. Resistive forces
oppose motion, not necessarily
velocity axes.

9709-27-C2

ANSWER

FagusED · A-Level 9709
4.4 Newton's laws of motion

Resultant force equals mass times
acceleration for a constant-mass
body.

9709-27-C1

ANSWER

FagusED · A-Level 9709
4.5 Energy, work and power

Loss in GPE= $mg\Delta h=100\text{ J}$;
 $\frac{1}{2}mv^2=100$; $v=10\text{ m s}^{-1}$.

9709-28-P

ANSWER

FagusED · A-Level 9709
4.5 Energy, work and power

Power is work per second or
instantaneous Fv when force acts
along velocity.

9709-28-C2

ANSWER

FagusED · A-Level 9709
4.5 Energy, work and power

Work by a force is component
along displacement times
distance. Net work changes
kinetic energy; gravitational
potential and elastic stores can
simplify a mechanics problem
when losses are absent.

9709-28-C1

ANSWER

FagusED · A-Level 9709
5.1 Representation of data

$12/3=4$; $18/6=3$.

9709-29-P

ANSWER

FagusED · A-Level 9709
5.1 Representation of data

Cumulative frequency estimates
median and percentiles; mean of
grouped data uses midpoints and
is approximate. Standard
deviation measures spread about
the mean, unlike range which
relies on extremes.

9709-29-C2

ANSWER

FagusED · A-Level 9709
5.1 Representation of data

Choose displays suited to data
type; histogram area, not height,
represents frequency if widths
differ.

9709-29-C1

ANSWER

FagusED · A-Level 9709
5.2 Permutations and combinations

Five positions with two L and two
E; $5!/(2!2!)=30$.

9709-30-P

ANSWER

FagusED · A-Level 9709
5.2 Permutations and combinations

For arrangements with repeated
identical objects divide by
repeated factorials. Restrictions
such as adjacency are often
handled by blocks or
complement.

9709-30-C2

ANSWER

FagusED · A-Level 9709
5.2 Permutations and combinations

Permutations count ordered
selections $nPr=n!/(n-r)!$;
combinations count unordered
groups $nCr=n!/[r!(n-r)!]$.

9709-30-C1

ANSWER

FagusED · A-Level 9709
5.3 Probability

Blue-green: $(3/5)(2/4)=3/10$;
green-blue: $(2/5)(3/4)=3/10$; total
 $3/5$.

9709-31-P

ANSWER

FagusED · A-Level 9709
5.3 Probability

Tree branches must use updated
probabilities after sampling
without replacement; check totals
sum to one.

9709-31-C2

ANSWER

FagusED · A-Level 9709
5.3 Probability

Conditional probability $P(A \text{ given } B)=P(A \cap B)/P(B)$. Independent
events satisfy $P(A \cap B)=P(A)P(B)$,
which differs from mutually
exclusive nonempty events.

9709-31-C1

ANSWER

FagusED · A-Level 9709
5.4 Discrete random variables

What is the central idea in Discrete random variables?

9709-32-C1

QUESTION

FagusED · A-Level 9709
5.4 Discrete random variables

Which mechanism, condition or distinction matters in Discrete random variables?

9709-32-C2

QUESTION

FagusED · A-Level 9709
5.4 Discrete random variables

For $P(X=0)=0.2$, $P(X=1)=0.5$, $P(X=2)=0.3$, find $E(X)$ and $\text{Var}(X)$.

9709-32-P

QUESTION

FagusED · A-Level 9709
5.5 The normal distribution

What is the central idea in The normal distribution?

9709-33-C1

QUESTION

FagusED · A-Level 9709
5.5 The normal distribution

Which mechanism, condition or distinction matters in The normal distribution?

9709-33-C2

QUESTION

FagusED · A-Level 9709
5.5 The normal distribution

For $X \sim N(50, 10^2)$, express $P(40 < X < 60)$ in standard normal form and estimate it.

9709-33-P

QUESTION

FagusED · A-Level 9709
6.1 The Poisson distribution

What is the central idea in The Poisson distribution?

9709-34-C1

QUESTION

FagusED · A-Level 9709
6.1 The Poisson distribution

Which mechanism, condition or distinction matters in The Poisson distribution?

9709-34-C2

QUESTION

FagusED · A-Level 9709
6.1 The Poisson distribution

Calls arrive at mean rate 3 per hour. Find $P(\text{no calls in 20 minutes})$.

9709-34-P

QUESTION

FagusED · A-Level 9709
6.2 Linear combinations of random variables

What is the central idea in Linear combinations of random variables?

9709-35-C1

QUESTION

FagusED · A-Level 9709
6.2 Linear combinations of random variables

Which mechanism, condition or distinction matters in Linear combinations of random variables?

9709-35-C2

QUESTION

FagusED · A-Level 9709
6.2 Linear combinations of random variables

Independent X, Y have $E(X)=3$, $E(Y)=5$, $\text{Var}(X)=2$, $\text{Var}(Y)=4$. Find $E(2X-Y)$ and $\text{Var}(2X-Y)$.

9709-35-P

QUESTION

FagusED · A-Level 9709
6.3 Continuous random variables

What is the central idea in Continuous random variables?

9709-36-C1

QUESTION

FagusED · A-Level 9709
6.3 Continuous random variables

Which mechanism, condition or distinction matters in Continuous random variables?

9709-36-C2

QUESTION

FagusED · A-Level 9709
6.3 Continuous random variables

For $f(x)=kx^2$ on $0 \leq x \leq 1$, find k and $P(X > 0.5)$.

9709-36-P

QUESTION

FagusED · A-Level 9709
5.4 Discrete random variables

$$E=0+0.5+0.6=1.1;$$
$$E(X^2)=0.5+1.2=1.7;$$
$$\text{Var}=1.7-1.1^2=0.49.$$

9709-32-P

ANSWER

FagusED · A-Level 9709
5.4 Discrete random variables

A binomial model requires fixed number of independent trials with constant success probability and two outcomes per trial.

9709-32-C2

ANSWER

FagusED · A-Level 9709
5.4 Discrete random variables

A probability distribution has nonnegative probabilities summing to 1. Expected value is $\sum xP(X=x)$; variance is $\sum x^2P(X=x) - [E(X)]^2$.

9709-32-C1

ANSWER

FagusED · A-Level 9709
5.5 The normal distribution

z bounds -1 and $+1$; probability $\Phi(1) - \Phi(-1) \approx 0.6827$.

9709-33-P

ANSWER

FagusED · A-Level 9709
5.5 The normal distribution

A continuous distribution has $P(X=a)=0$. The normal approximation to a discrete model needs suitable conditions and continuity correction where applicable.

9709-33-C2

ANSWER

FagusED · A-Level 9709
5.5 The normal distribution

Normal distributions are symmetric with mean μ and standard deviation σ ; standardise $z=(x-\mu)/\sigma$ before using tables.

9709-33-C1

ANSWER

FagusED · A-Level 9709
6.1 The Poisson distribution

Mean for $1/3$ h is $\lambda=1$;
 $P(X=0)=e^{-1} 1^0/0! \approx 0.3679$.

9709-34-P

ANSWER

FagusED · A-Level 9709
6.1 The Poisson distribution

For twice the exposure, double λ ; a binomial approximation may apply for large n and small p with $\lambda=np$.

9709-34-C2

ANSWER

FagusED · A-Level 9709
6.1 The Poisson distribution

Poisson models counts of events in a fixed interval under independent, constant-rate assumptions. $P(X=r)=e^{-\lambda} \lambda^r/r!$, with mean and variance λ .

9709-34-C1

ANSWER

FagusED · A-Level 9709
6.2 Linear combinations of random variables

$$E=2(3)-5=1; \text{Var}=4(2)+4=12.$$

9709-35-P

ANSWER

FagusED · A-Level 9709
6.2 Linear combinations of random variables

Variances are not subtracted when finding difference of independent variables.

9709-35-C2

ANSWER

FagusED · A-Level 9709
6.2 Linear combinations of random variables

Linearity gives $E(aX+bY)=aE(X)+bE(Y)$. For independent X, Y , $\text{Var}(aX+bY)=a^2\text{Var}(X)+b^2\text{Var}(Y)$; covariance changes this if dependence exists.

9709-35-C1

ANSWER

FagusED · A-Level 9709
6.3 Continuous random variables

Normalise $k \int_0^1 x^2 dx = k/3 = 1$, so $k=3$; $P(X>0.5)=1-(0.5)^3=0.875$.

9709-36-P

ANSWER

FagusED · A-Level 9709
6.3 Continuous random variables

A cumulative distribution $F(x)$ integrates f and is nondecreasing from 0 to 1.

9709-36-C2

ANSWER

FagusED · A-Level 9709
6.3 Continuous random variables

For continuous density f , probability between bounds is area under curve. Normalisation $\int f=1$ fixes constants; $E(X)=\int xf(x)dx$ and $\text{Var}(X)=\int x^2f(x)dx - E(X)^2$.

9709-36-C1

ANSWER

FagusED · A-Level 9709
6.4 Sampling and estimation

What is the central idea in Sampling and estimation?

9709-37-C1

QUESTION

FagusED · A-Level 9709
6.4 Sampling and estimation

Which mechanism, condition or distinction matters in Sampling and estimation?

9709-37-C2

QUESTION

FagusED · A-Level 9709
6.4 Sampling and estimation

A normal population has known $\sigma=12$; a random sample $n=36$ has mean 50. Form approximate 95% z interval for μ .

9709-37-P

QUESTION

FagusED · A-Level 9709
6.5 Hypothesis tests

What is the central idea in Hypothesis tests?

9709-38-C1

QUESTION

FagusED · A-Level 9709
6.5 Hypothesis tests

Which mechanism, condition or distinction matters in Hypothesis tests?

9709-38-C2

QUESTION

FagusED · A-Level 9709
6.5 Hypothesis tests

A right-tail z test yields $p=0.032$. State decisions at 5% and 1% and explain what a 5% significance level controls.

9709-38-P

QUESTION

FagusED · A-Level 9709
6.4 Sampling and estimation

$SE=12/\sqrt{36}=2$;
margin= $1.96 \times 2=3.92$; interval
(46.08,53.92).

9709-37-P

ANSWER

FagusED · A-Level 9709
6.4 Sampling and estimation

A confidence interval combines
estimate with critical value times
standard error; distinguish
confidence of method from
probability of a fixed parameter.

9709-37-C2

ANSWER

FagusED · A-Level 9709
6.4 Sampling and estimation

A statistic varies from sample to
sample; its sampling distribution
describes that variation. For
independent observations from
suitable population, sample mean
is unbiased for μ with
 $\text{Var}(\bar{X})=\sigma^2/n$.

9709-37-C1

ANSWER

FagusED · A-Level 9709
6.5 Hypothesis tests

Reject H_0 at 5%, fail to reject at
1%; 5% is maximum long-run
type I error probability under H_0 .

9709-38-P

ANSWER

FagusED · A-Level 9709
6.5 Hypothesis tests

Compare p-value with
significance level or statistic with
critical region; final conclusion
must be in context and must not
claim H_0 proven. A one-sided test
requires a justified directional
claim made in advance.

9709-38-C2

ANSWER

FagusED · A-Level 9709
6.5 Hypothesis tests

State H_0 and H_1 with parameter
and direction before calculating a
test statistic.

9709-38-C1

ANSWER